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# From sonochemical synthesis to predictive modeling: Unraveling the antioxidant properties of La-doped $CeO_2$ nanoparticles

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ABSTRACT

This study introduces a novel combination of sonochemical synthesis and machine learning (ML) modeling to analyze the effect of Lanthanum (La) doping on the biological properties of cerium oxide ($CeO_2$) nanoparticles with various La concentrations (0, 1, 5, and 10 at.%). Ultrasonic-assisted synthesis enabled La incorporation into the $CeO_2$ lattice, resulting changes in crystallinity, lattice parameters, and surface features. Detailed characterization confirmed successful doping and indicated stable nanoparticle suspensions with controlled size ranges. Sonochemical synthesis promoted the oxidation of $Ce^{3+}$ to $Ce^{4+}$. Biological testing revealed low cytotoxicity across various cell lines (HepG-2, 3 T3-L1, Caco-2, and U87) and increased antioxidant activity, especially in samples with 5 and 10 at.% La, which demonstrated improved free radical scavenging of DPPH and $H_2O_2$ radicals. Notably, advanced ML models—including Extremely Randomized Trees, random forest, Gradient Boosting, and LightGBM—enabled the prediction of antioxidant activity based on nanoparticle features, identifying antioxidant method, concentration, and chemical composition as key factors influencing biological effects. This combined experimental and data-driven approach not only clarifies the structure–activity relationships of La-doped $CeO_2$ nanoparticles but also emphasizes the significant potential of ML in designing and optimizing nanomaterials for biomedical applications. The combination of sonochemical synthesis and ML modeling provides a robust framework for accelerating nanomaterial development by minimizing trial-and-error experiments and providing mechanistic insights into their biological functions.

## 1. Introduction

Use of lanthanide-doped materials in biomedicine is attracting considerable interest due to their distinctive antioxidant properties and biocompatibility [1–4]. Among these materials, cerium oxide ($CeO_2$) nanoparticles, commonly called nanoceria, are promising tools for biomedicine due to their antioxidant properties [5,6]. The most prevalent dopants for $CeO_2$ include gadolinium (Gd) [7,8], europium (Eu) [9,10], and samarium (Sm) [11,12]. Doped $CeO_2$ materials are crucial for advancing diagnostic and therapeutic technologies in the biomedical sector. These nanoparticles mimic the function of antioxidant enzymes, such as superoxide dismutase (SOD) and catalase, and thus may be used to treat conditions associated with oxidative damage, including neurodegenerative diseases, cancer, and inflammatory

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disorders. Their low toxicity, favorable biocompatibility, and ability to penetrate cells make them suitable for various biomedical applications [13,14].

Lanthanum ($La^{3+}$) is a silvery-white lanthanide. Compared with other elements from the same category, $La^{3+}$ offers advantages of malleability, high stability at high temperatures, high electrical conductivity, high density, and useful biomedical properties (e.g., biocompatibility, low toxicity, and ability to stimulate cell proliferation and differentiation) [15,16]. Despite its useful properties, $La^{3+}$ and related salts have been reported to cause neurotoxicity and impair learning and memory by overactivating N-methyl-D-aspartate (NMDA) receptors in neurons [17]. In addition to challenges related to ease of oxidation, corrosion, and cost, $La^{3+}$ and derived salts have been reported to cause aberrations in motor function, energy metabolism, and synaptic plasticity [18]. In the development of bioactive nanostructures, $La^{3+}$ or its stable forms have been used to decorate the surface of carbon-dot composites loaded with camptothecin for breast cancer treatment [19], synthesize functionalized mesoporous nanoplatforms for in vitro and in vivo treatment of malignant glioblastoma [20], and develop spherical materials for surgical procedures (e.g., bone implantation) [21]. Lanthanum-derived nanomaterials, such as chitosan-coated siRNA-loaded lanthanum phosphate nanoparticles, were reported to inhibit tumor growth in a chemically induced orthotopic colorectal cancer mouse model [22]. It has also been described to initiate reorganization of the cytoskeleton in cervical carcinoma cell lines and deregulate mitochondrial function in uterine squamous carcinoma cell lines [23]. In other cancer cell models, lanthanum-based nanocrystals have been documented to induce pyroptosis in lung cancer cells by increasing reactive oxygen species (ROS) levels and mimicking the activity of catalase and glutathione peroxidase. [22]. Although the biocompatibility of La has been reported, studies related to La-doped $CeO_2$ have focused on the photocatalytic and magnetic performance of these materials [23,24]. Regarding their synthesis, the sonochemical method is a facile approach to producing nanostructured $CeO_2$ and composites with desirable morphologies and enhanced surface area [25–28].

Sonochemical synthesis offers an environmentally friendly, efficient method of producing metal nanoparticles via ultrasound-induced acoustic cavitation [29,30]. This process creates localized hot spots with extremely high temperatures and pressures, promoting rapid nucleation and growth of nanoparticles under gentle conditions, such as room temperature and atmospheric pressure [31]. The sonochemical approach supports direct reduction pathways without requiring harsh conditions or dispersants, enabling precise control over particle size, shape, and composition [32]. Sonochemical methods have been successfully used to synthesize various nanomaterials, including metal nanoparticles (e.g., Au, Ag, Ni), metal oxides (e.g., $TiO_2$, $Fe_3O_4$, CuO), metal chalcogenides (e.g., $MoS_2$, CuS, SnS, $SnS_2$), alloy nanoparticles, and heterojunction nanocomposites [33]. For example, sonochemical co-reduction has produced Au-Ag core–shell nanoparticles, iron colloids with narrow size distributions, $TiO_2$ and CuO nanoparticles with improved photocatalytic properties, and copper sulfide nanoparticles created in a single step with higher reaction efficiency than solvothermal methods [34,35]. Sonochemical synthesis has also been used to prepare quantum dots with tunable optical bandgaps, such as SnS and $SnS_2$, which demonstrate superior photocatalytic and sonocatalytic activity [36]. The versatility and efficiency of sonochemical synthesis make it a state-of-the-art technique for creating nanomaterials with properties tailored to use in catalysis, environmental cleanup, biomedicine, and more.

Machine learning (ML) is accelerating our understanding of material properties, particularly of complex nanomaterials such as nanoparticles. By analyzing experimental datasets, ML algorithms can identify patterns and correlations that may not be identified immediately using traditional methods [37,38]. ML models can predict how variations in synthesis conditions, doping elements, or structural features affect the biological activity of nanoparticles. Such predictive insights enable the efficient design and development of $CeO_2$-based materials for biomedical applications, eliminating the time and financial costs associated with experimental trial-and-error approaches [39]. Modeling complex, nonlinear phenomena in data science necessitates advanced techniques that capture a range of patterns in data. Such patterns can be analyzed using supervised ML models. In supervised ML, algorithms are trained on labeled datasets [40,41]. A model learns to map the input features to known output labels so it can make predictions on new, unseen data. The learning process involves minimizing the difference between the predicted and actual outputs through iterative adjustments. Supervised learning is commonly used for classification and regression in applications ranging from image recognition to numerical prediction. It is a powerful tool for solving well-defined problems with clear target variables, as its effectiveness relies on the quality and quantity of the labeled training data [42,43].

This work introduces a novel combination of sonochemical synthesis and machine learning (ML) modeling to explore the biomedical properties of La-doped $CeO_2$ nanoparticles at different doping levels. Unlike previous studies that mainly focused on photocatalytic and magnetic properties, this research thoroughly investigates the structural, morphological, and surface features of these nanoparticles while assessing their cytotoxicity and antioxidant activity using various radical assays. The application of advanced ML techniques, such as random forest, Extremely Randomized Trees, Gradient Boosting, and LightGBM, to predict antioxidant activity based on physicochemical attributes like zeta potential, carboxyl groups, and La content offers new insights into the structure–activity relationships of this nanomaterial. This combined experimental and data-driven approach enhances our understanding of how La doping influences antioxidant behavior, emphasizing the potential of these nanomaterials for biomedical applications and demonstrating the significant role of ML in the design and optimization of nanomaterials.

## 2. Experimental section

### 2.1. Nanoparticle synthesis

La-doped $CeO_2$ (0, 1, 5, and 10 at.%) nanoparticles were prepared using an ultrasonic-assisted method. For this purpose, 200 mL of deionized water was heated at 80 °C under stirring. Then, 0.2 g of polyvinyl alcohol (Mw: 70000–100000, 87 %–90 % hydrolyzed) was added. Stoichiometric amounts of $Ce(NO_3)_3{\cdot}6H_2O$ and $La(NO_3)_3{\cdot}6H_2O$ were incorporated into the obtained transparent solutions. The samples were subjected to ultrasonic treatment for 15 min, separated into intervals of 40–20 s in an on/off cycle (amplitude, 70 %; Q700 sonicator, 20 kHz, Qsonica). The solutions were evaporated under continuous stirring and dried at 200 °C for 3 h. The final thermal treatment, drying at 400 °C for 3 h, was applied. The samples were labeled CL0, CL1, CL5, and CL10 according to their La content.

### 2.2. Material characterization

X-ray diffraction (XRD) patterns were obtained using an Empyrean diffractometer (PANalytical, $\lambda_{Cu}$ = 1.5406 Å). Attenuated total reflectance–Fourier transform infrared (ATR–FTIR) spectra were recorded in the 4000–400 $cm^{-1}$ range using an IR Affinity-1S spectrometer (Shimadzu, Columbia, MD, USA). Ultraviolet–visible–near-infrared (UV–Vis-NIR) spectra were obtained using a Cary 5000 spectrometer (Agilent Technologies). The hydrodynamic particle size (DLS) and zeta potential were measured with a Zetasizer Pro analyzer (Malvern Instruments). Field-Emission Scanning Electron Microscopy (FE-SEM) was performed in a Gemini 3 instrument (Zeiss).Transmission electron microscopy (TEM) and scanning TEM (STEM) were performed using a JEOL ARM-200F (Laboratorio Universitario de Microscopía Electrónica, LUME@UNAM, RRID: SCR_024400). X-ray photoelectron spectroscopy was carried out with a K-Alpha System (Thermo Scientific). The high-

resolution XPS (HR-XPS) spectra of Ce 4d, O 1 s, C 1 s, and La 4d were obtained at a pass energy of 15 eV and a step size of 0.1 eV. The XPS spectra were deconvoluted using the software AAnalyzer and aligned to the C 1 s signal at 284.8 eV [44].

### 2.3. Biological activity

#### 2.3.1. Cell culture and cytotoxicity assay

The cytotoxicity of the La-doped $CeO_2$ nanomaterials was tested against hepatocellular, colon, and glioblastoma cell lines. The cytotoxicity of the obtained nanomaterials was investigated against fibroblasts. All cell lines were cultured in DMEM F-12 medium under previously established conditions. The MTT assay was used to determine the cytotoxicity of the La-doped $CeO_2$ nanomaterials and performed as described in [4].

#### 2.3.2. Antioxidant assays

The antioxidant performance of the La-doped $CeO_2$ nanomaterials was assessed using the DPPH, ABTS, and $H_2O_2$ assays. For the DPPH assay, a stock solution was placed in a 96-well plate, mixed with La-doped $CeO_2$ nanomaterials, and preserved in a dark environment for 30 min. For the ABTS assay, a stock solution was first prepared with potassium persulfate and then mixed with La-doped $CeO_2$ nanomaterials at 5, 20, 50, 100, and 250 μg/mL, respectively. The $H_2O_2$ assay was executed by dispensing 70 mL of a stock solution into a 96-well plate and mixing it with the La-doped $CeO_2$ nanomaterials. Ascorbic acid (AC) was employed as a positive control. For all the antioxidant assays, the absorbance of treatment was assessed as previously published, considering the wavelengths and preservation period required for each technique [4].

### 2.4. Statistical analysis

The experimental data were analyzed using ANOVA and a post hoc Tukey's honestly significant difference test with 95 % confidence. Statistical significance is indicated using asterisks according to the p-value: *$p \leq 0.05$, **$p \leq 0.01$, or *** $p \leq 0.001$.

### 2.5. ML modeling

*Dataset*: The dataset consisted of 180 observations, each containing 22 attributes and one response variable (**Table S1**). The attributes corresponded to variables such as crystal size, lattice parameters, bandgap, zeta potential, areas of various bonds, dopant content (Ce and La), and sample concentration. The response was the antioxidant activity. All variables except two were numeric (Material and Method).

*Modeling*: The typical procedure for transitioning from a dataset to trained and validated models in ML generally commences with data collection and preparation. This phase encompasses data cleaning (addresses missing values and outliers), variable transformation, and dataset division into training and test sets. When categorial values are present, it is convenient to transform them using a technique called "One-Hot Encoding". Conventionally, approximately 70 %–80 % of the data are allocated for training, and the remainder are reserved for testing. Subsequently, pertinent features are selected, and data exploration is conducted to discern patterns and relationships. Given adequately prepared data, an ML algorithm appropriate for a specific problem (e.g., regression, classification, or clustering) is chosen. The model is then trained using the training dataset, and the hyperparameters are adjusted to optimize model performance. Subsequently, the model is validated using the test dataset to evaluate its accuracy, sensitivity, specificity, and other relevant metrics. In certain instances, cross-validation can be used to ensure that the model effectively generalizes to new data. Ultimately, the model is deployed for production if it satisfies the anticipated performance criteria.

For the next phase of experiments, supervised ML models were built using the attributes in the dataset (i.e., the 22 independent variables), with antioxidant activity as the dependent variable (i.e., the variable being predicted). Given that two of those variables are categorical, when OHE transformation is applied, the dataset has 26 independent variables. In the first stage, four models were considered as baseline models: random forest (RF), Extremely Randomized Trees (EXT), XGBoost (XGB), and LightGBM (LGBM). An RF utilizes an ensemble of decision trees to enhance model robustness and accuracy by averaging multiple predictions, thereby reducing the risk of overfitting. Extremely Randomized Trees is an ensemble learning method. It builds multiple trees and makes predictions based on averaging the outputs of the individual trees. Extra Trees selects splits more randomly than a random forest. Specifically, for each candidate feature, a split point is randomly selected from the feature's value range. XGB and LXGB are boosting techniques that build sequential models, each correcting the errors of the previous one, achieving high accuracy and computational efficiency. These techniques are particularly useful for investigating complex phenomena because they effectively handle nonlinear relationships and large datasets. A schematic representation of the ML experiment is shown in Scheme 1.

The most promising model was identified from this initial set of experiments, and its hyperparameters were optimized efficiently by randomly selecting a set of hyperparameter values within a predefined space, a strategy called random search. Unlike a grid search, a random search does not exhaustively explore all possible combinations. Instead, it randomly explores the hyperparameter space, enabling the discovery of good configurations without incurring prohibitive computational costs. Because it facilitates the identification of hyperparameter configurations that offer good performance in a short time, a random search is useful in cases with large hyperparameter spaces and limited time and computational resources.

*Modeling performance*: Various metrics are used to evaluate the accuracy and effectiveness of ML models, such as the mean squared error (MSE), root MSE (RMSE), mean absolute error (MAE), and coefficient of determination (R2). The MSE measures the average of the squares of errors, or the differences between the predicted and actual values, providing a clear indication of the error's magnitude regardless of its sign. The RMSE is the square root of the MSE, and its unit is the same as that of the original variable, facilitating interpretation. R2 evaluates the proportion of the total variation in the dependent variable that the model can explain, providing a measure of the model's explanatory power. $R^2$ values close to 1 indicate a good model–data fit. Finally, the MAE is another metric commonly used to assess the accuracy of ML models. Unlike the MSE and RMSE, the MAE measures the average absolute error, or the absolute difference between the predicted and actual values, without squaring them. The MAE provides a direct measure of the average magnitude of errors, so it is easier to interpret and is less sensitive to outliers than the MSE and RMSE [45].

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (1)$$

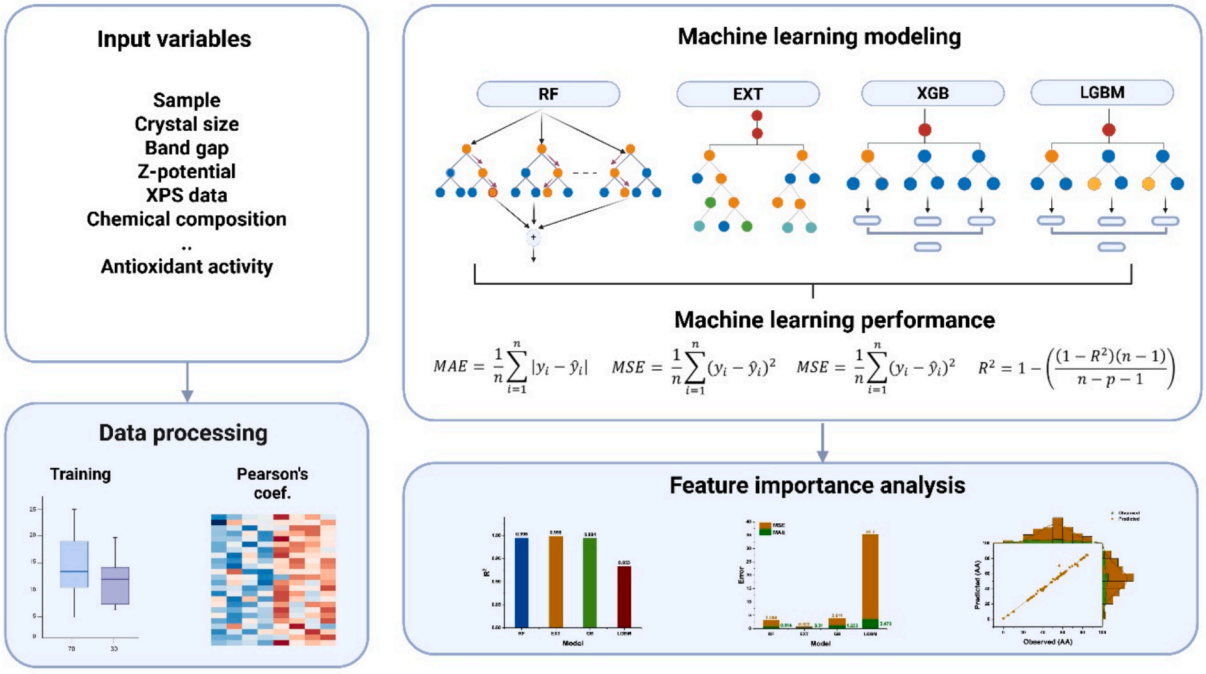
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (2)$$

$$VS = 1 - \frac{var(y - \hat{y})}{var(y)} \quad (3)$$

## 3. Results and discussion

### 3.1. Material characterization

The XRD patterns of the $Ce_{1-x}La_xO_2$ (x = 0, 1, 5, 10 at.%) nanoparticles are shown in Fig. 1a. The patterns correspond to $CeO_2$ of the JCPDS #34–0394 card, and no impurities or phases related to doping




**Scheme 1.** Workflow of ML modeling.

were detected, indicating the incorporation of $La^{3+}$ ions within the $CeO_2$. The incorporation of $La^{3+}$ ions reduced crystallinity and changed the 2(θ) position slightly. These effects were evident in the normalized XRD intensity (Fig. 1b). The crystallite size and structural parameter *a* (cubic structure) were calculated from the XRD patterns (Fig. 1c). The slight variation of the *a* value was attributed to the ionic radii of $Ce^{4+}$ (0.97 Å) and $La^{3+}$ (1.16 Å) [46]. The ATR-FTIR spectra of the materials (Fig. 1d) exhibit different bands at 3340 $cm^{-1}$ (–OH), 2347 $cm^{-1}$ (atmospheric $CO_2$), and 1638 $cm^{-1}$ (H–O–H). Other bands at 1523 and 1360 $cm^{-1}$ are related to C=O and N–O, respectively. The 1049 and 770 $cm^{-1}$ peaks corresponded to C–H stretching and bending modes, respectively. The band at 861 $cm^{-1}$ corresponds to O–Ce–O bonding. The general stretching of bands to higher wavenumbers confirmed the incorporation of $La^{3+}$ into $CeO_2$.

Fig. 2a demonstrates the absorbance spectra of the La-doped $CeO_2$ nanoparticles in the 200–1000 nm range. The samples exhibited an adsorption band at 240 nm, corresponding to the electronic transition from the 2p state to 4f of O and Ce. The materials did not exhibit variations in absorbance, and no additional absorption bands were observed after the incorporation of the $La^{3+}$ ions. The Kubelka–Munk function was derived from these absorbance spectra to determine the optical bandgap (Eg) values [47] for CL0, CL1, CL5, and CL10 to be 3.10, 3.11, 3.09, and 3.09 eV, respectively. The $E_g$ values were used to determine the valence band and conduction band positions at the point of zero charge, along with the redox potential of reactive oxygen species (ROS) (Fig. 2b). The hydrodynamic diameters of the nanoparticles are shown in Fig. 2c. The samples exhibited a bimodal diameter, with the first mode being localized at approximately 160 ± 5 nm. The second modal values, which showed significant variation, corresponded to 1227, 1267, 936, and 1861 ± 5 % nm for CL0, CL1, CL5, and CL10, respectively. The zeta potential values were 39.95, 28.20, 32.78, and 28.14 ± 5 % mV for CL0, CL1, CL5, and CL10, respectively, indicating a stable suspension.

An illustration of the synthesis route is provided in Fig. 3a. Sonochemical-assisted synthesis alters material morphology through physical and chemical mechanisms. Initially, cavitation bubbles are generated during the nucleation phase, which evolve to result in either stable or transient cavitation. Transient cavitation takes place when bubbles expand beyond their equilibrium size and then collapse. This event produces shock waves and shear forces, which are extremely energetic. This was verified by analyzing the influence of ultrasound on the morphology of the La-doped $CeO_2$ materials via field emission scanning electron microscopy (FESEM; Figs. 3b–3e). These images showed similar morphologies across all samples. Their morphologies consisted of large material segments of interconnected, collapsed bubbles, displaying high porosity. No secondary morphologies were observed after doping, indicating the dominance of sonochemical effects. **Fig. S1** shows the EDS spectra of the samples.

High-resolution TEM (HRTEM) analysis was conducted on CL0 (Fig. 4a) and CL10 (Fig. 4b) to further investigate the FESEM-observed surface characteristics of the morphologies. The low-magnification images of both materials indicated that the FESEM-observed surface morphologies were entirely nanostructured. The average particle size was under 20 nm. From the selected-area electron diffraction (SAED) images, the (111)–(220) and (111)–(311) planes were ascribed to CLO and CL10, respectively. Two sections in the high-magnification images were analyzed. Fast Fourier transform (FFT) was applied, and inverse FFT (IFFT) was used to obtain the lattice fringes, which belonged to the (1 1 1) and (200) crystal planes. For CL0, the (1 1 1) facet spacing was 3.09 Å, and the (200) facet was 2.64 Å. For CL10, the (1 1 1) facet was 3.13 Å, and the (200) facet was 2.63 Å. These changes corroborated the variation in the XRD-obtained cell parameters. For CL10, STEM-EDX analysis (Fig. 4c) was conducted to observe element distributions (Ce, O, and La). A STEM-EDX line analysis of this sample indicated a uniform element distribution (Fig. 4d).

XPS measurements were performed on the La-doped $CeO_2$ nanoparticles (x = 0, 1, 5, 10 at.%). The survey spectra (**Fig. S2**) of pure $CeO_2$ and the La-doped $CeO_2$ nanoparticles demonstrate that the main elements observed were Ce, O, and C. As the La concentration increased, peaks appeared at approximately 100.6 eV (La 4d) and 835 eV (La 3d). The most probable XPS signals of Ce and La were the 3d core levels between 825 and 955 eV. However, these signals appeared at high binding energies, with strong contributions from the background signal owing to electron scattering. Due to the complexity of these core levels, analysis should focus on the 4d levels, which exhibited high sensitivity in the photoemission signal. **Fig. S3** illustrates the C 1 s spectra, which consisted of three chemical components related to the C–C (284.8 eV),

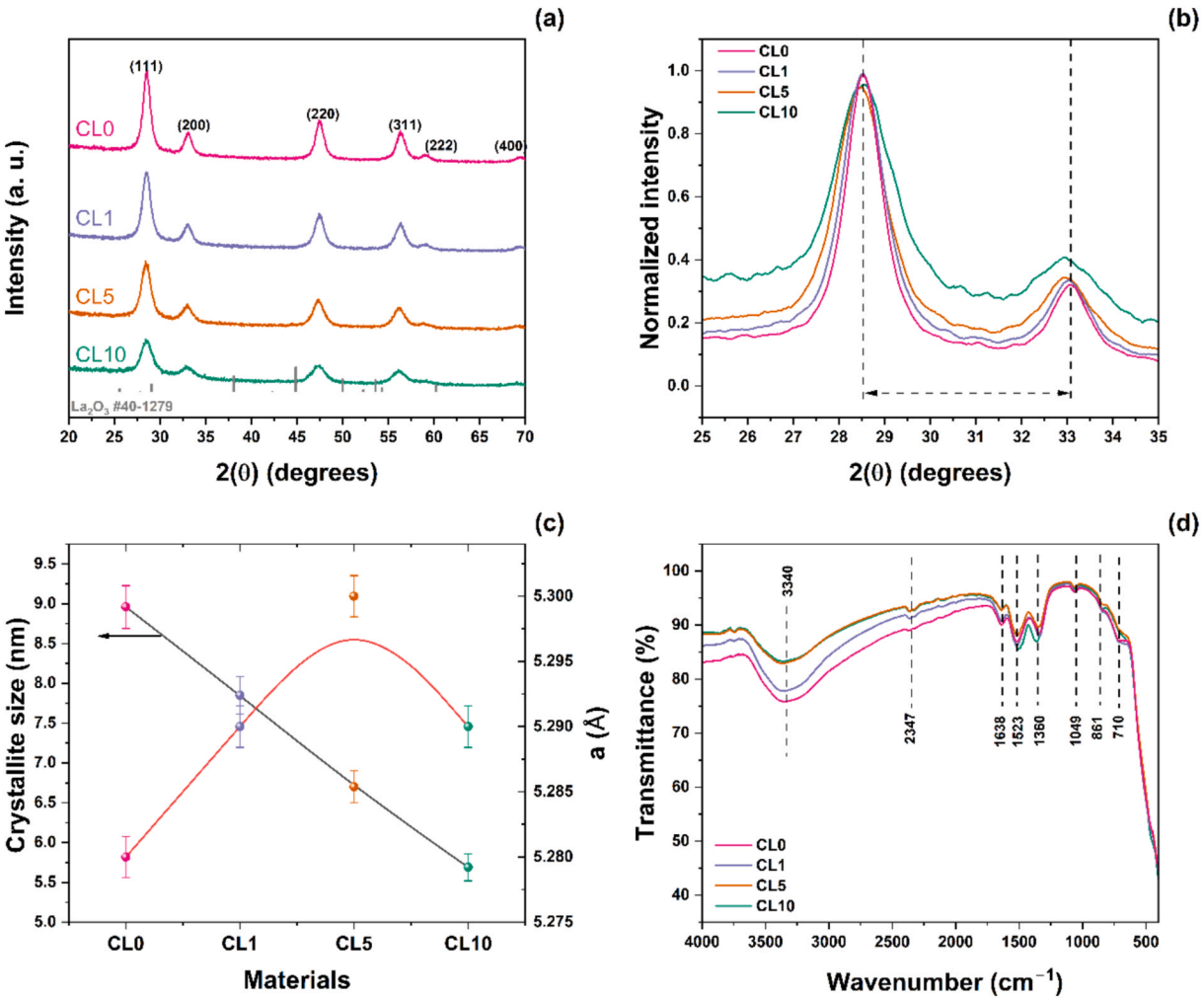



**Fig. 1.** Structural characterization of the nanoparticles. (a) XRD patterns, (b) normalized intensity and magnification of XRD patterns, (c) average crystallite size and *a*-axis length, and (d) ATR-FTIR spectra of La-doped $CeO_2$ nanoparticles.

C–O (286.1 eV), and C=O (288.9 eV) bonds. These adventitious carbon signals arose from the exposure of the samples to the environment. In the O 1 s spectra, two peaks are observed, which are associated with $O^{2-}$ ions in the $CeO_2$ and OH– with centers at 528.9.0 and 531.1 eV, respectively (Fig. 5a) [39]. The Ce and La signals were close in binding energy, so analysis was focused on the XPS spectra at 90–180 eV, especially the extremes, to reproduce their background adequately (Fig. 5b). The complex electronic structure of rare-earth ions causes difficulty in adjusting spectral shapes and discriminating the origin of each chemical component. Therefore, this study utilized information obtained from theoretical calculations predicting final electronic structures after photoemission [48]. In particular, the Ce 4d region was fitted to five chemical components. Voigt-type profiles were used for the main components of the Ce 4d spectra, whereas Gaussian profiles were used for the peaks at high binding energies. For low binding energies, three doublet peaks labeled C1, C2, and C3, with spin–orbit separations of 3.1, 4.5, and 3.27 eV, respectively, were used. The C1 chemical component (107.8 eV) was related to the $Ce^{4+}$ ions in the $CeO_2$ crystal structure, whereas the C2 (113.8 eV) and C3 (121.3 eV) chemical components were associated with satellite peaks [49]. The singlet peaks at 138.5 and 150.9 eV were associated with energy loss. As the La concentration increased in the $CeO_2$ nanoparticles, a doublet peak at 100.9 eV with 2.8 eV spin–orbit separation appeared, which was related to the $La^{3+}$ ions in the $CeO_2$ crystal structure. Therefore, the XPS analysis results demonstrated the successful doping of $CeO_2$ nanoparticles with La ions. $CeO_2$ prepared via wet chemical methods almost always has a $Ce^{3+}/Ce^{4+}$ ratio [50]. However, as observed in the XPS results, only $Ce^{4+}$ was present in all samples (dry condition). This could have been due to the decomposition of water into hydroxyl and perhydroxyl radicals by the sonochemical waves [51]. These radicals, which have oxidant properties, could have enabled the oxidation of $Ce^{3+}$ to $Ce^{4+}$.

*Cytotoxicity assay*: The cytotoxicity of CL0, CL1, CL5, and CL10 was evaluated in the 3 T3-L1, HepG-2, and Caco-2 cell lines. The 3 T3-L1 cells are derived from a Swiss albino mouse embryonic model. The HepG-2 cell line is a representative model of hepatocellular carcinoma, whereas Caco-2 is widely used for investigating molecular and cellular events related to human colon adenocarcinoma.

### 3.2. Biological activity

At initial concentrations (5, 20, and 50 μg/mL), CL0 did not cause cytotoxic effects in the 3 T3-L1 cells, whose viability remained comparable to that of the untreated control (Fig. 6). However, at higher concentrations, treatment resulted in 94.34 % ± 0.81 % and 87.91 % ± 2.78 % cell viability, respectively. Furthermore, 5–100 μg/mL of CL1 did not affect the viability of the 3 T3-L1 cells, but treatment at 250 μg/mL resulted in 88.23 % ± 0.99 % cell viability. Treatment with 5–50 μg/mL of CL1 exerted similar effects (no noted changes in cell viability). However, after treatment with 100 and 250 μg/mL of CL1, cell viability was 97.41 % ± 2.29 % and 91.06 % ± 0.92 %, respectively. Among the developed materials, CL10 exhibited the highest cytotoxicity toward the 3 T3-L1 cells; treatment at 5, 20, and 50 μg/mL resulted in 95.23 % ±

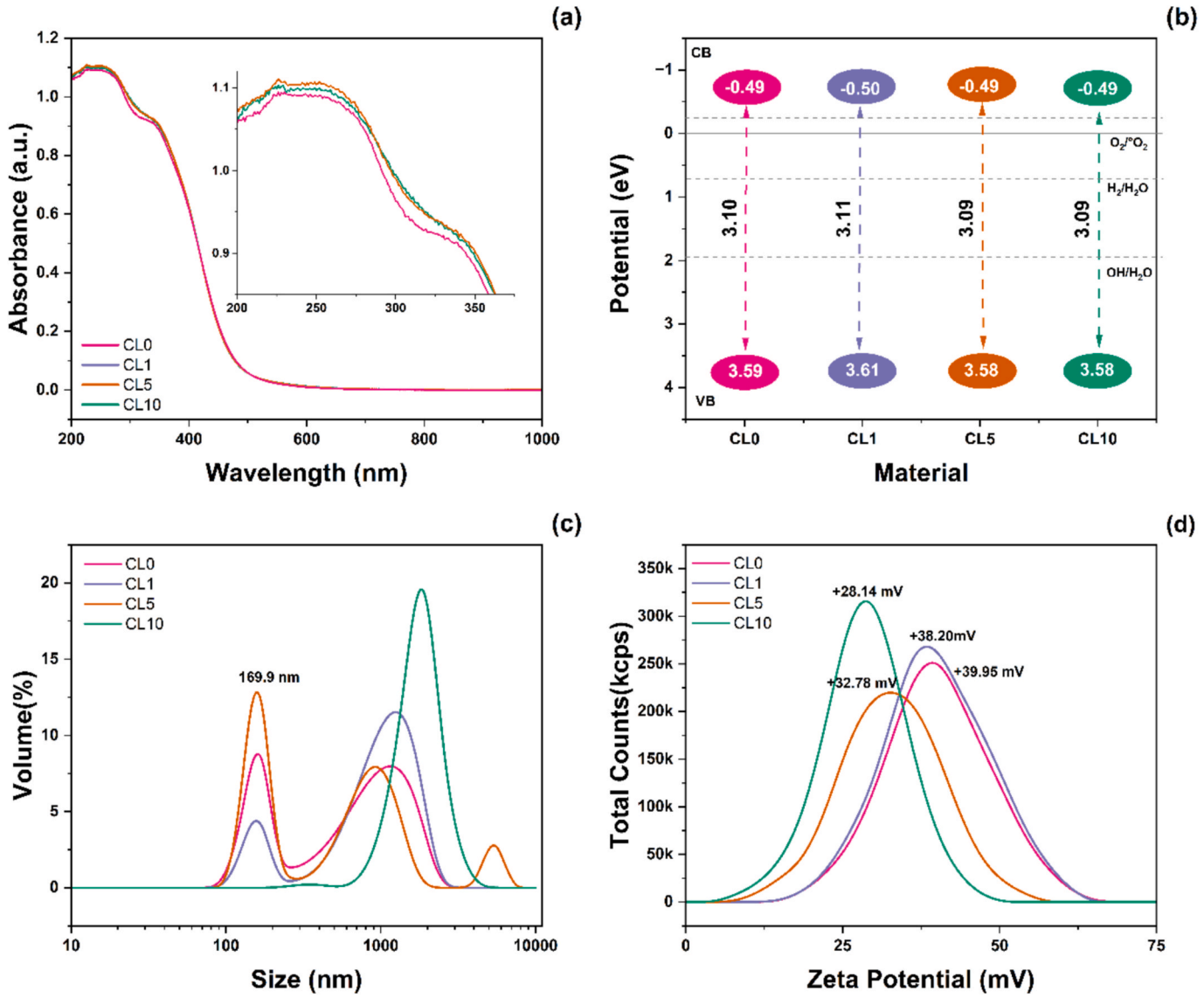



**Fig. 2.** Spectroscopic characterization of the nanoparticles. (a) UV–vis absorption spectra, (b) optical bandgap, (c) size distribution analysis of DLS, and (d) zeta potentials of La-doped $CeO_2$ materials.

5.83 %, 94.31 % ± 2.62 %, and 94.72 % ± 1.32 % cell viability, respectively. At 100 and 250 μg/mL, treatment resulted in 92.17 % ± 1.82 % and 88.63 % ± 1.33 % cell viability, respectively. Additional parameters, such as the half-maximal lethal concentration, were not calculated because CL0, CL1, CL5, and CL10 did not compromise the viability of the cultured cell lines. In fact, at certain concentrations, treatment with nanoparticles enhanced cell proliferation, as observed in Figs. 6a, c, and d with 250 μg/mL of CL1, CL5, and CL10.

*Antioxidant activity*: The antioxidant activity of CL0, CL1, CL5, and CL10 was tested using DPPH, ABTS, and $H_2O_2$ radicals. The DPPH and ABTS assays are colorimetric methods that leverage the ability of organic or inorganic antioxidants to donate electrons to neutralize free radical formation. Similarly, the $H_2O_2$ assay yields essential information about the antioxidant potential of substances and is used in drug development to determine the potential of organic and inorganic molecules to mitigate oxidative stress; prevent diseases; modulate cell signaling; and reverse protein, lipid, and DNA damage. Experimentally, the scavenging capacity of antioxidants has been compared with that of other synthetic or natural antioxidants, such as AC, butylated hydroxytoluene, and quercetin. AC consists of a six-carbon lactone ring and is widely implemented to reduce inflammation, enhance lymphocyte function, stimulate cytokine production, and diminish the detrimental effects of oxidative stress in biological systems.

*DPPH assay*: Treatment at 5, 20, and 50 μg/mL inhibited 41.12 %, 58.58 %, and 60.45 % of DPPH, respectively (Fig. 7a). Similarly, treatment with 100 and 250 μg/mL of CL0 scavenged 60.51 % and 63.43 % of the radicals, respectively. Furthermore, treatment with 5 and 20 μg/mL of CL1 inhibited the formation of 54.05 % and 54.42 % of DPPH, whereas treatment at 50 and 100 μg/mL scavenged 56.23 and 57.92 % of the radicals, respectively. Treatment with 250 μg/mL of CL1 inhibited 58.13 % of DPPH. Then, 5–50 μg/mL of CL5 inhibited 59.07 %–59.85 % of DPPH, whereas treatment at 100 and 250 μg/mL scavenged 60.93 % and 61.40 % of the free radicals, respectively. Treatment at 5, 20, and 50 μg/mL scavenged 57.18 %, 59.07 %, and 60.27 % of DPPH, respectively. Treatment at higher concentrations (100 and 250 μg/mL) inhibited the formation of 60.93 % and 62.12 % of the radicals, respectively. Furthermore, treatment with 5, 20, and 50 μg/mL of AC scavenged 82.36 %, 86.75 %, and 90.84 % of the radicals, respectively. At 100 and 250 μg/mL, treatment with AC inhibited 91.49 % and 92.30 % of DPPH, respectively. The $IC_{50}$ values of the nanoparticles' activity against DPPH are listed in Table 1.

*$H_2O_2$ assay*: Treatment with 5 μg/mL of CL0 inhibited 38.41 % of

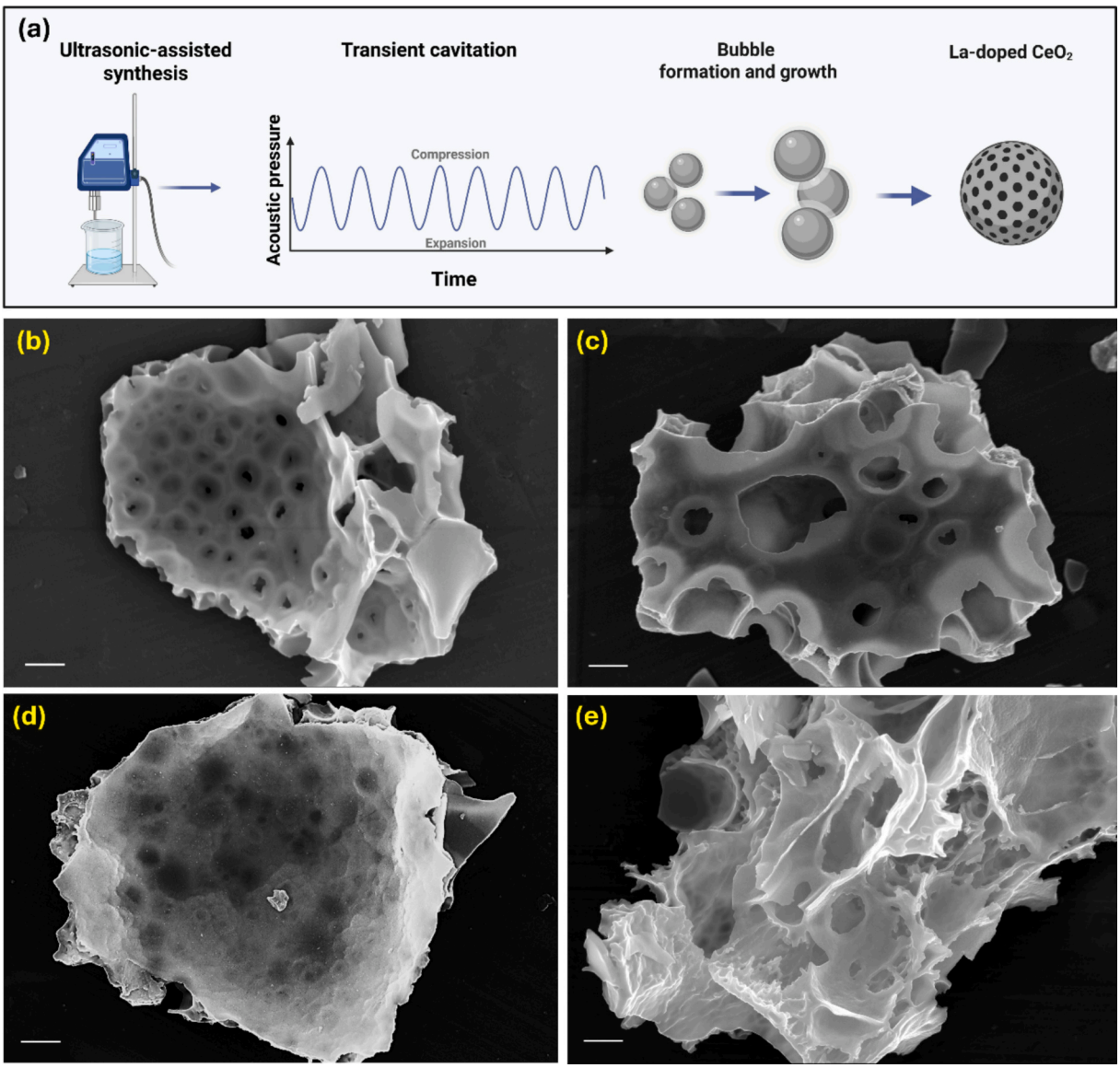



**Fig. 3.** Morphological results of the nanoparticles. (a) Schematic representation of the synthesis process and (b–e) low-magnification FESEM images of CL0, CL1, CL5, and CL10. Scale bar = 1 μm.

$H_2O_2$, whereas treatment at 20, 50, and 100 μg/mL scavenged 40.81 %–40.82 % of the free radicals (Fig. 7b). Treatment with 250 μg/mL of CL10 inhibited 40.92 % of $H_2O_2$. By contrast, treatment at 5, 20, and 50 μg/mL scavenged 48.90 %, 52.15 %, and 53.15 % of $H_2O_2$, respectively. At sample concentrations of 100 and 250 μg/mL, treatment with CL1 decreased 55.18 % and 56.45 % of the available radicals. In the same assay, 5 and 20 μg/mL of CL5 scavenged 66.71 % and 69.50 % of the free radicals, whereas at 50 and 100 μg/mL, it scavenged 74.15 % and 74.29 % of $H_2O_2$, respectively. Treatment with 250 μg/mL of CL5 inhibited 75.12 % of the radicals. Regarding CL10 performance, 5, 20, and 50 μg/mL resulted in the scavenging of 78.85 %, 80.85 %, and 81.54 % of $H_2O_2$, respectively. With 100 and 250 μg/mL of CL10, the generation of 84.37 % and 84.70 % of the free radicals, respectively, was inhibited. Similar to CL10, treatment with 5 and 20 μg/mL of AC scavenged 70.03 % and 77.14 % of $H_2O_2$, whereas treatment at 50 and 100 μg/mL inhibited 85.95 % and 96.17 % of free radicals, respectively. Treatment with 250 μg/mL of AC inhibited 97.72 % of $H_2O_2$. Table 1 presents the $IC_{50}$ values of CL0, CL1, CL5, and CL10 for $H_2O_2$.

*ABTS assay:* Treatment with 5 and 20 μg/mL of CL10 scavenged 10.25 % and 24.45 % of ABTS; at 50, 100, and 250 μg/mL, CL0 limited the production of 27.63 %, 28.83 %, and 30.54 % of ABTS, respectively (Fig. 7c). It was determined that 5, 20, and 50 μg/mL of CL1 hindered the formation of 4.67 %, 25.47 %, and 26.39 % of ABTS, respectively. Remarkably, at 100 and 250 μg/mL, treatment with CL1 scavenged 36.39 % and 40.06 % of ABTS, respectively. As with CL1, treatment with 5 and 20 μg/mL of CL5 inhibited 33.21 %, and 38.30 % of ABTS, whereas at 50 and 100 μg/mL it hampered the development of 38.48 % and 39.59 % of the free radicals, respectively. When analyzed at higher concentrations, 250 μg/mL of CL5 obstructed the formation of 39.80 % of ABTS. Among the developed nanoparticles, CL10 had the highest ABTS-scavenging capacity: at 5, 20, and 50 μg/mL, treatment inhibited 42.39 %, 43.18 %, and 48.09 % of ABTS, respectively. At 100 and 250 μg/mL, treatment with CL10 inhibited 72.81 % and 78.14 % of ABTS, respectively. Regarding the AC results, 5, 20, and 50 μg/mL resulted in the inhibition of 95.27 %, 95.61 %, and 96.15 % of ABTS, respectively. Finally, it was determined that 100 and 250 μg/mL of AC hindered the formation of 99.44 % and 99.48 % of free radicals, respectively. The $IC_{50}$ of the nanoparticles' activity against ABTS is listed in Table 1.

The antioxidant activities of the La-doped $CeO_2$ nanoparticles against the DPPH, ABTS, and $H_2O_2$ radicals were classified according to

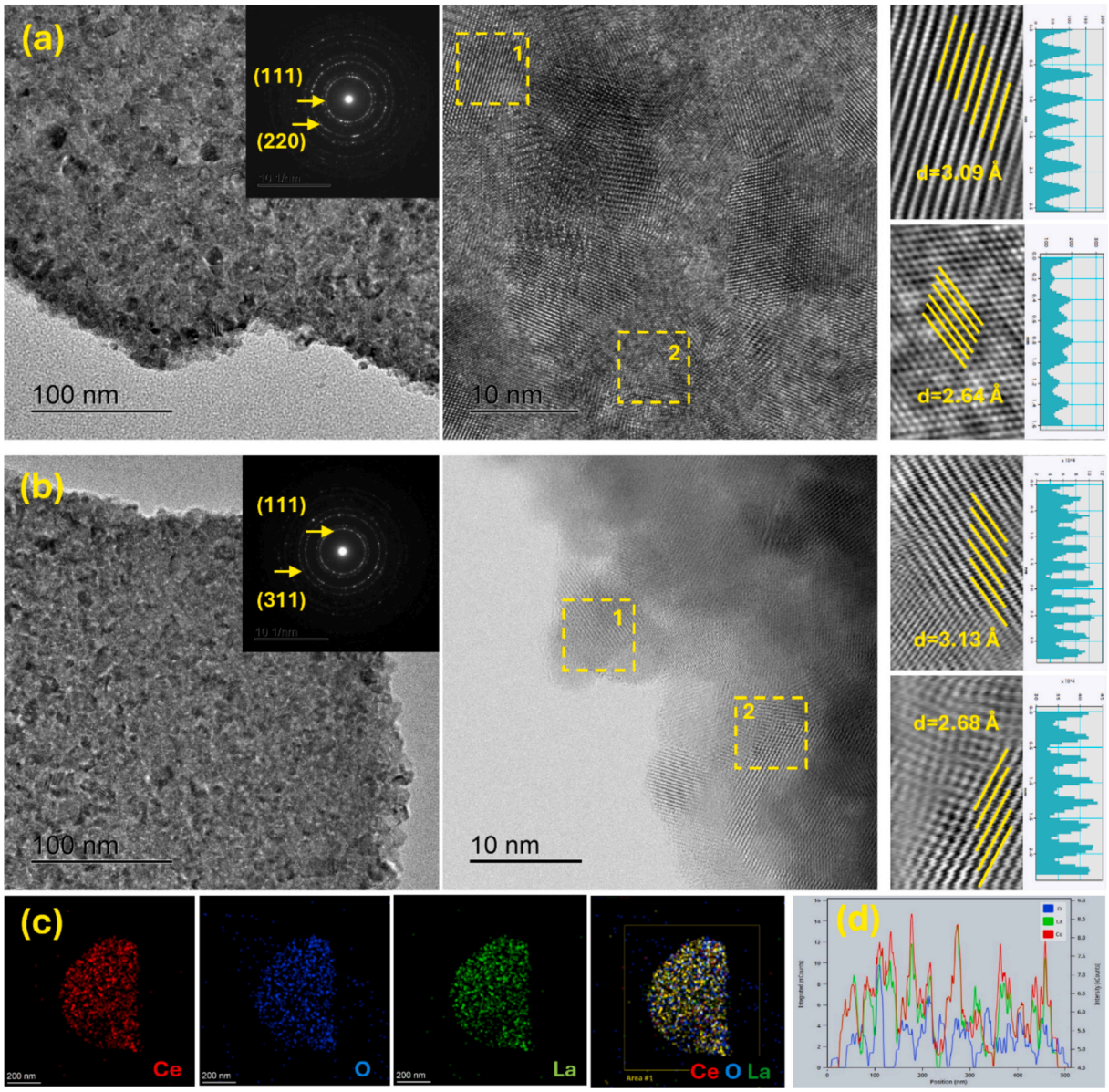



**Fig. 4.** TEM and HRTEM images, SAED patterns, and IFFT images of (a) CL0 and (b) CL10. (c) STEM-EDX analysis and (d) composition profile of CL10.

their $IC_{50}$ values. In the DPPH assay, the antioxidant performance of CL0 was moderate, whereas that of CL1, CL5, and CL10 was high. Moreover, CL0 and CL5 exhibited low scavenging activities, whereas those of CL1 and CL10 were moderate. In the $H_2O_2$ assay, the antioxidant activities of CL10 and CL1 were weak and moderate, respectively, whereas CL5 and CL10 demonstrated scavenging capacity. The antioxidant activities of AC in the DPPH, ABTS, and $H_2O_2$ assays were high. CL5 and CL10 exhibited superior scavenging abilities in the DPPH and $H_2O_2$ assays, as indicated by their $IC_{50}$ values. The high antioxidant activity of CL1, CL5, and CL10 could be crucial for their potential use in treating Alzheimer's and Parkinson's diseases and neoplasticity-related disorders.

The antioxidant activity of the La-doped $CeO_2$ nanoparticles was attributed to their SOD mimetic activity, which stems from the intrinsic reversible + 3 and + 4 oxidation states of cerium and is associated with oxidative stress mitigation, improved cellular function, and enhanced cellular protection. The capacity of the $CeO_2$ nanoparticles to mimic SOD activity enables them to catalyze the conversion of hazardous chemical species, such as $H_2O_2$, into oxygen and water (Fig. 7d). In their reduced form (3 + ), $CeO_2$ nanoparticles can utilize oxygen atoms and transform them into water. Free radicals are inhibited due to interactions between oxidizing and reducing agents, where the former gains electrons and the latter donates them [52]. In the inhibition of free radical formation during treatment with inorganic nanoparticles, such as nanoceria, the roles of oxidizing and reducing agents can vary according to the specific reaction and the nature of the nanoparticles involved [53]. Here, the tested free radicals acted as oxidizing agents, whereas the developed materials served as reducing agents, facilitating the $Ce^{3+} \leftrightarrow Ce^{4+}$ cycle.

### 3.3. ML modeling

#### 3.3.1. Correlation analysis

In this study, correlation analysis was conducted on the provided variables. Correlation analysis should be conducted between variables before ML modeling to help identify relationships within a dataset. This step is important for several reasons. High correlation between variables may indicate redundancy, suggesting they convey identical information. Eliminating such redundancies simplifies the model and reduces the risk of overfitting. Avoiding highly correlated variables can enhance model accuracy and efficiency, allowing ML algorithms to focus on relevant

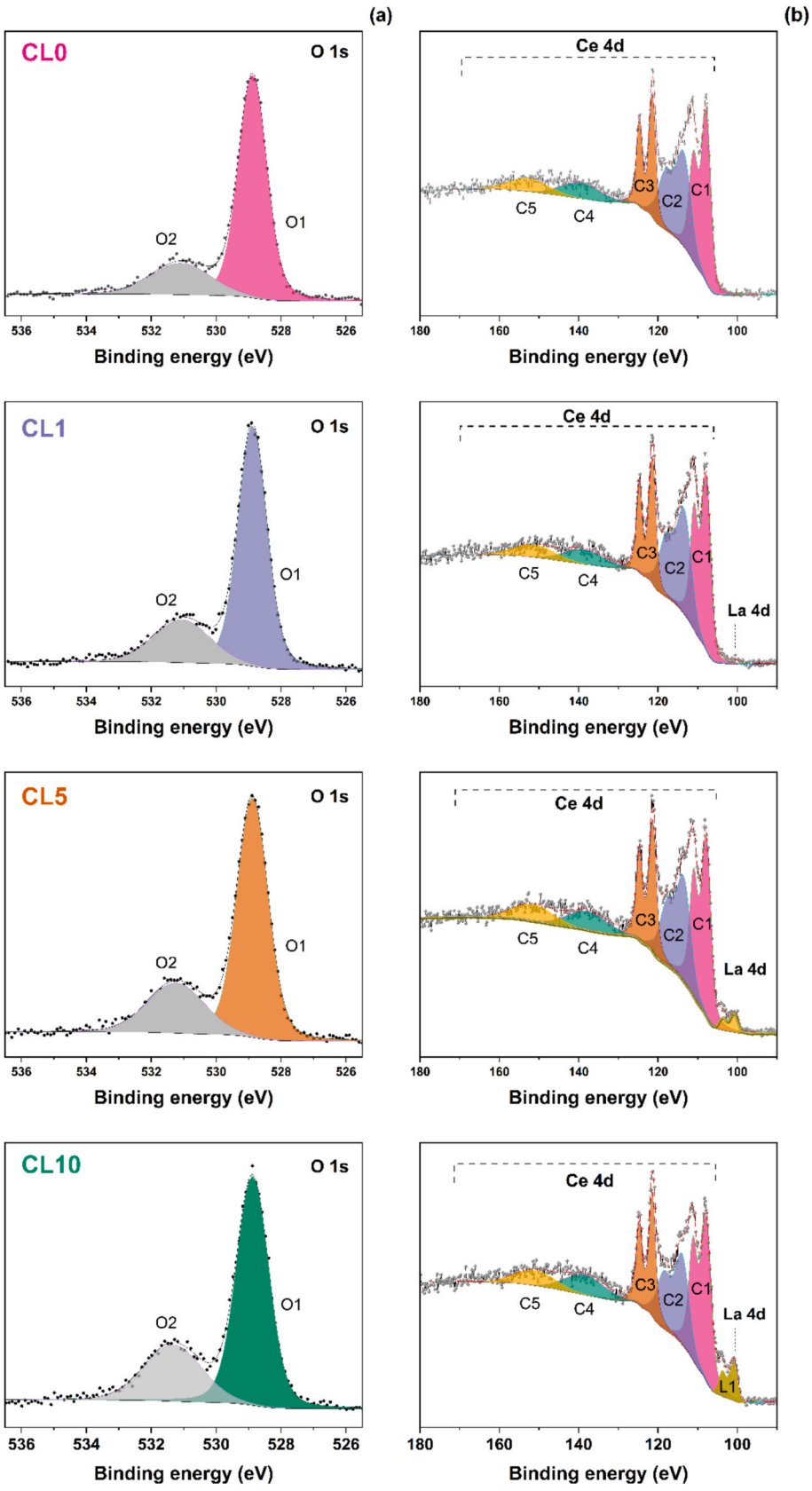



**Fig. 5.** HR-XPS spectra of (a) O 1 s and (b) Ce 4d–La 4d.

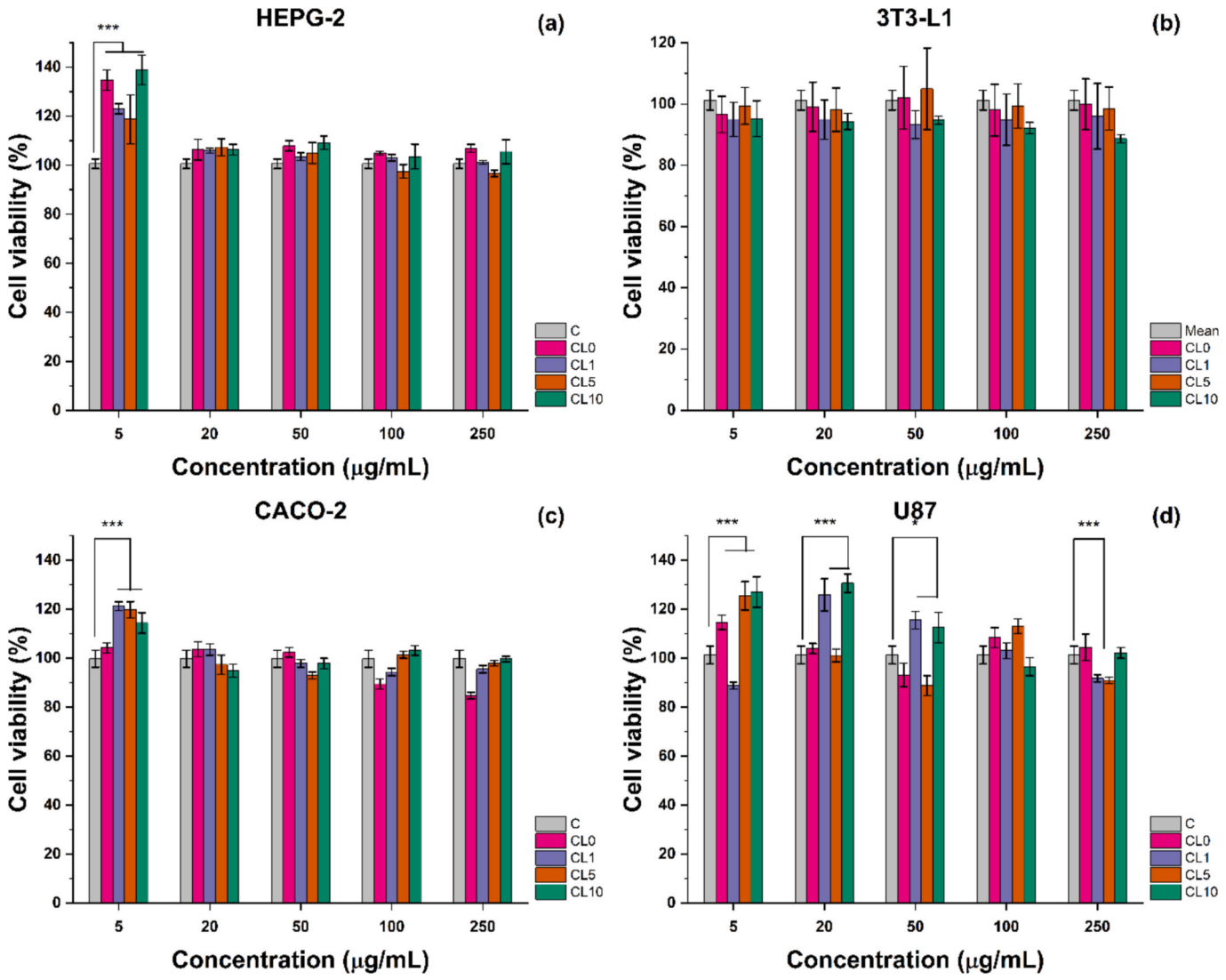



**Fig. 6.** Cytotoxicity of La-doped $CeO_2$ nanoparticles toward HepG-2 (a), 3 T3-L1 (b), Caco-2 (c), and U87 (d). The mean ± standard deviation (SD) of three independent experiments is shown. *$p < 0.05$, ** $p < 0.01$, and ***$p < 0.001$.

variables without being hindered by redundant information. Additionally, correlation analysis can reveal variable relationships that may be vital to model performance, thereby guiding feature selection and variable engineering for modeling optimization. In this work, Pearson's correlation analysis did not reveal a linear relationship. However, it suggested that variables such as molecular weight (MW) and material type (CL10) significantly affected antioxidant activity. This is illustrated in Fig. 8, which displays the correlation index value per variable pair.

The variables were reviewed to detect possible correlations and then standardized to homogenize their values. Variable standardization is crucial for data preprocessing in ML modeling. It involves transforming variables such that they have a mean of zero and an SD of one. This is particularly important when using algorithms that are sensitive to data scales, such as logistic regression, neural networks, and distance-based methods (e.g., K-nearest neighbor). Standardization helps improve model convergence during training and prevents large-scale variables from dominating the learning process, improving the balance and effectiveness of model performance. After this preprocessing, unsupervised analysis, called clustering, can be performed. These methods attempt to find patterns in data by grouping them according to similarity. Several methods can be used for such grouping. In this study, the K-means method was applied, considering the possible existence of two clusters. This process involves utilizing all attributes to create a multidimensional model. A dimension reduction technique called principal component analysis (PCA) was applied to visualize the results.

### 3.3.2. *Unsupervised ML: Clustering*

Using two clusters, K-means clustering was applied to determine the possibility of distinguishing between cases with and without antioxidant activity. After the clusters were calculated, PCA was adopted to reduce the model to two dimensions (pca1 and pca2) for plotting. The result is provided in **Fig. S4**, which shows four separate blobs, two per category. This outcome can be interpreted in several ways. First, it suggests that the data may have a more complex structure than the two initial clusters. The four separate blobs imply the presence of two well-defined subclusters within each initial cluster. This indicates that the original data contained more than two natural clusters, but the K-means algorithm, which was applied with only two clusters, failed to capture this complexity. PCA reduced the data dimensionality to two principal components, which accounted for the majority of the variability in the dataset. The clear separation into four blobs suggests that this variability can be represented well in these two dimensions, indicating the suitability of PCA for visualizing these data. The distinct separation could also suggest that the data within each blob are quite heterogeneous and different from one another, indicating the presence of distinctive features in the original data that led to this segmentation.

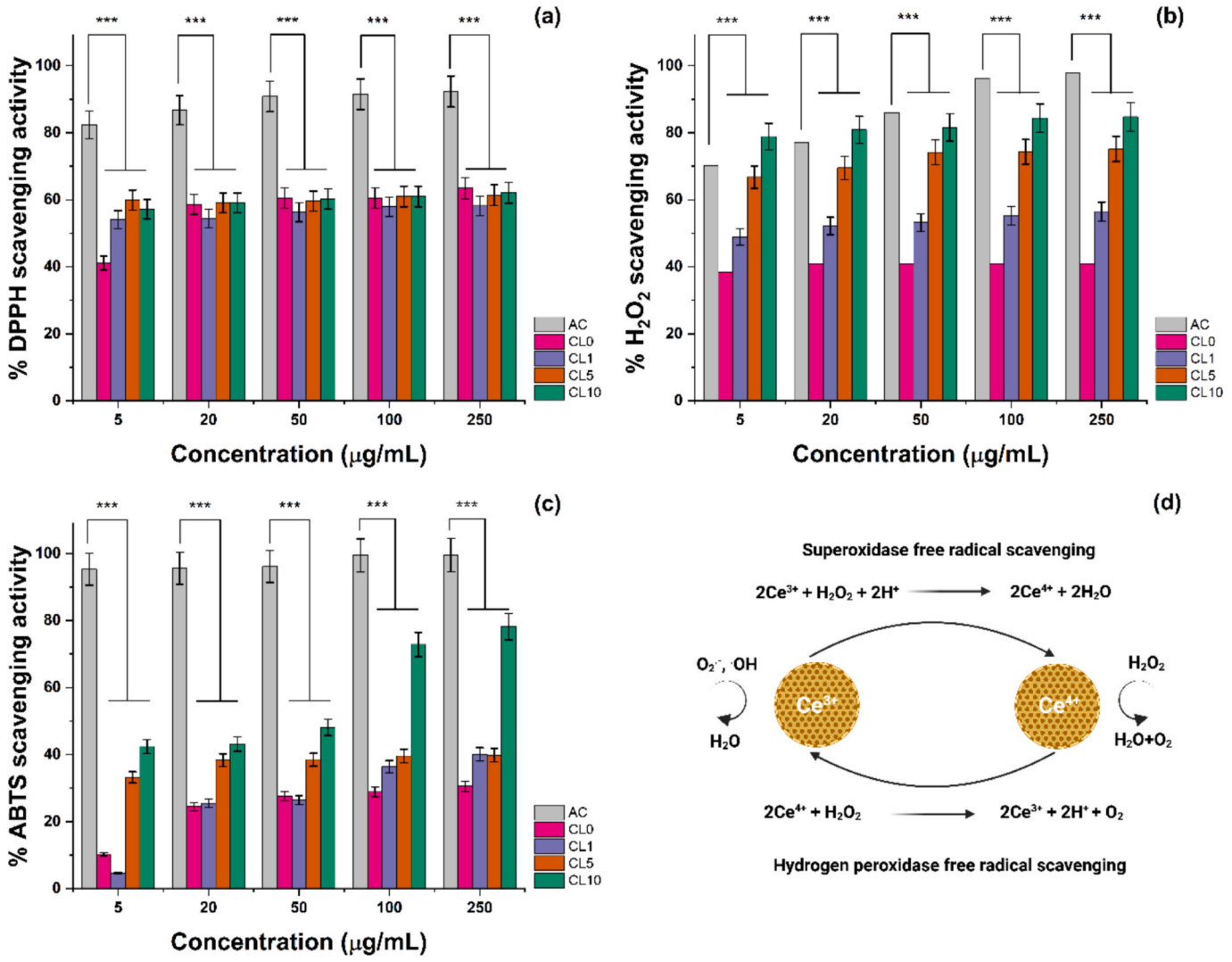



**Fig. 7.** Antioxidant activity of La-doped $CeO_2$ nanoparticles against (a) DPPH, (b) $H_2O_2$, and (c) ABTS. The mechanism by which the synthesized nanoparticles can scavenge free radicals is depicted in (d). The mean ± SD of three independent experiments is shown. *$p < 0.05$, ** $p < 0.01$, and ***$p < 0.001$.

**Table 1**
$IC_{50}$ values (μg/mL) of CL0, CL1, CL5, and CL10 for DPPH, ABTS, and $H_2O_2$.

| Sample | DPPH | ABTS | $H_2O_2$ |
|---|---|---|---|
| CL0 | 39.05 | 569.52 | 1904.52 |
| CL1 | 2.47 | 303.66 | 42.24 |
| CL5 | 3.05 | 798.05 | 1.89 |
| CL10 | 2.71 | 40.41 | 3.15 |
| AC | 3.07 | 3.40 | 2.42 |

### 3.3.3. *Supervised ML*

With the data already standardized, a feature selection analysis was conducted to evaluate the significance of the considered attributes. In machine learning, feature selection is the process of identifying and selecting the most relevant features or variables from a dataset to improve model performance. It is important because it reduces dimensionality, mitigates overfitting, enhances model interpretability, and decreases computational cost. By focusing on the most informative features, models can learn more effectively and generalize better to new data. The results of feature selection guide the construction of more efficient and accurate models by eliminating redundant or irrelevant data, ultimately leading to improved prediction accuracy and robustness in machine learning applications. In this work, feature selection was performed using the SelectKBest method in scikit-learn. For regression, it works by selecting the top k features that have the strongest relationship with the target variable, based on a specified scoring function. It is a filter-based feature selection technique that evaluates each feature individually and ranks them according to their statistical correlation or relevance to the target. The scoring function used was f_regression, which computes the correlation between each feature and the target using the F-test for linear regression. The process involves i) calculating the score for each feature using the chosen scoring function, ii) ranking features by their scores, and iii) selecting the top k features with the highest scores. For the dataset available, using SelectKBest with k = 10, the following features were determined to be the best: MW, MCN, Mce, Mla, AWCe, AWLa, CS, ZP, La, and Method_ABTS. Fig. 10 shows the scores for all attributes.

When performing feature selection, it is important to consider whether the change in the $R^2$ coefficient is the result of adding more predictors or features, regardless of their relevance to the model. To avoid this situation, an adjusted $R^2$ can be computed and reported. Adjusted R2 is a modified version of the $R^2$ that has been adjusted for the number of predictors in the model. While $R^2$ measures the proportion of variance in the dependent variable that can be predicted from the independent variables, Adjusted R2 adjusts this proportion based on the number of predictors and the sample size. The formula for Adjusted $R^2$ is

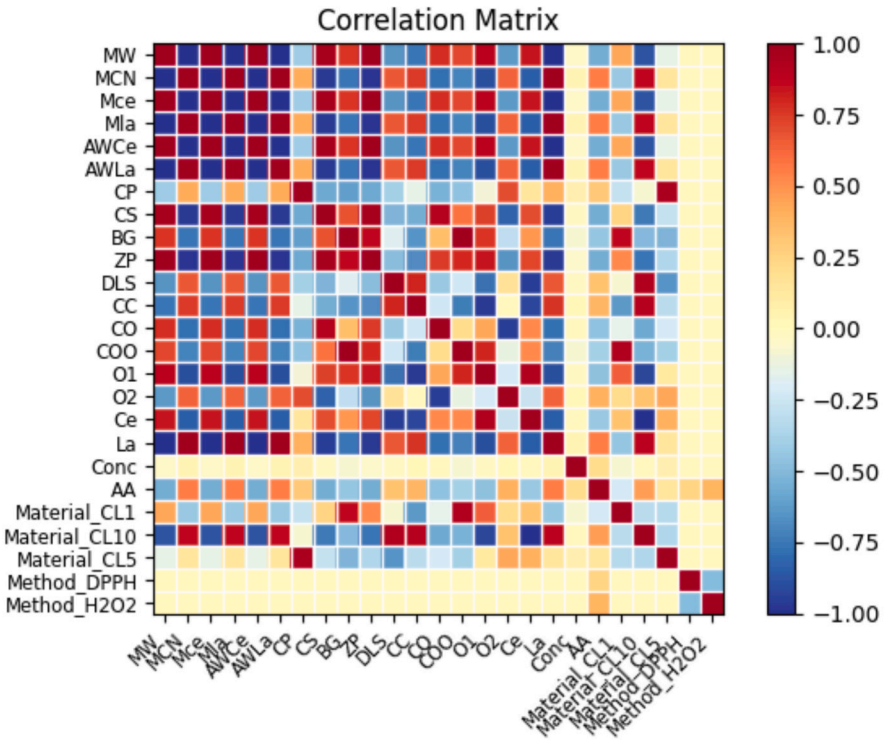



**Fig. 8.** Correlation matrix for features in the dataset. The numbers indicate the linear correlation between each feature pair. Zero means no correlation at all, and 1 or − 1 indicates absolute linear correlation.

[54]:

$$Adjusted R^2 = 1 - \left(\frac{(1-R^2)(n-1)}{n-p-1}\right) \quad (4)$$

where $R^2$ is the coefficient of determination, $n$ is the number of observations, and $p$ is the number of predictors (independent variables). Since Adjusted $R^2$ includes a penalty for adding unnecessary predictors, it only increases if the new predictor improves the model more than would be expected by chance. Thus, it provides a more accurate measure of model performance when multiple predictors are involved. Table 2 shows the results for the first five features included. There is a peak in the Adjusted $R^2$ value at 0.659944, which occurs with a certain number of features (likely around 8 features, based on the pattern). This suggests that this subset of features provides a good balance between model fit and complexity. After this peak, the Adjusted $R^2$ value plateaus or slightly decreases (e.g., an Adjusted $R^2$ of 0.651373 for the highest feature count), indicating that adding more features does not significantly improve the model and might even result in overfitting. The relatively high $R^2$ and Adjusted $R^2$ values towards the later rows (especially around the peak) indicate that the variables are relevant and contribute significantly to predicting the target outcome. The declining or stable Adjusted $R^2$ after the peak suggests that further variables do not add informative value and may not be relevant. For that reason, two models were ultimately tested: one fitted with all features and another with the eight variables shown. Table 2 presents the $R^2$ and Adjusted $R^2$ results for up to five features in the dataset. Each row incorporates one additional feature in order of decreasing F value. The variable with the highest importance is the method ABTS, followed by the zeta potential. The table containing all variable values is available in the supplementary materials (**Table S2**).

**Table 2**
$R^2$ and Adjusted $R^2$ for RF built with different numbers of features in the dataset. Each row describes the $R^2$ and Adjusted $R^2$ values for a model built with the indicated features, starting from the one with the highest importance.

| $R^2$ | Adj $R^2$ | Feature name | Feature |
|---|---|---|---|
| 0.2233 | 0.2178 | [ABTS] | 1 |
| 0.4049 | 0.5993 | [ZP, ABTS] | 2 |
| 0.5965 | 0.5878 | [MW, ZP, ABTS] | 3 |
| 0.6077 | 0.5964 | [MW, AWCe, ZP, ABTS] | 4 |
| 0.5955 | 0.5809 | [MW, MCe, AWCe, ZP, ABTS] | 5 |

Note: see **Table S1** in the Supplementary Materials for acronyms.

To establish a robust baseline, four different machine learning models were trained: random forest (RF), Extremely Randomized Trees (EXT), Gradient Boosting (GB), and Light Gradient Boosting (LGBM). Establishing a baseline is critical because it provides a reference point for gauging the performance of more complex models. By evaluating these diverse algorithms, we can uncover initial insights into the relationships within the data and identify the most promising models. The results from these baseline models inform subsequent steps in the modeling workflow, guiding hyperparameter tuning, feature engineering, and model selection for further refinement and optimization.

As previously noted, the four models are very similar in terms of their $R^2$ values, which range from 0.93 to 0.99. Fig. 9a shows the behavior of this parameter in the four models. Error is another important parameter to observe. In this case, Fig. 9b shows the behavior of the two reported errors, MAE and MSE, and it is possible to see a greater difference, with values ranging from 0.31 for MAE on EXT to 3.5 for LGBM. In the case of MSE, it ranges from 0.32 for EXT to 31.7 for LGBM.

Cross-validation was used to minimize bias in model selection. The results in Fig. 9c show that the EXT model achieved the minimum MAE ($1.163 \pm 0.5$), while LGBM had the worst MAE ($3.89 \pm 0.69$). Therefore, EXT was used for further analysis. We conducted a randomized search to find the best EXT model. The search was performed over a large parameter space of almost 40,000 combinations. The best estimator across all searched params was as follows: n_estimators = 100, min_samples_split = 2, min_samples_leaf = 1, max_leaf_nodes = 50, max_features = sqrt, max_depth = 10 and bootstrap = False. This optimized model resulted in a score of 0.999 and an MAE of 0.31. An error plot was

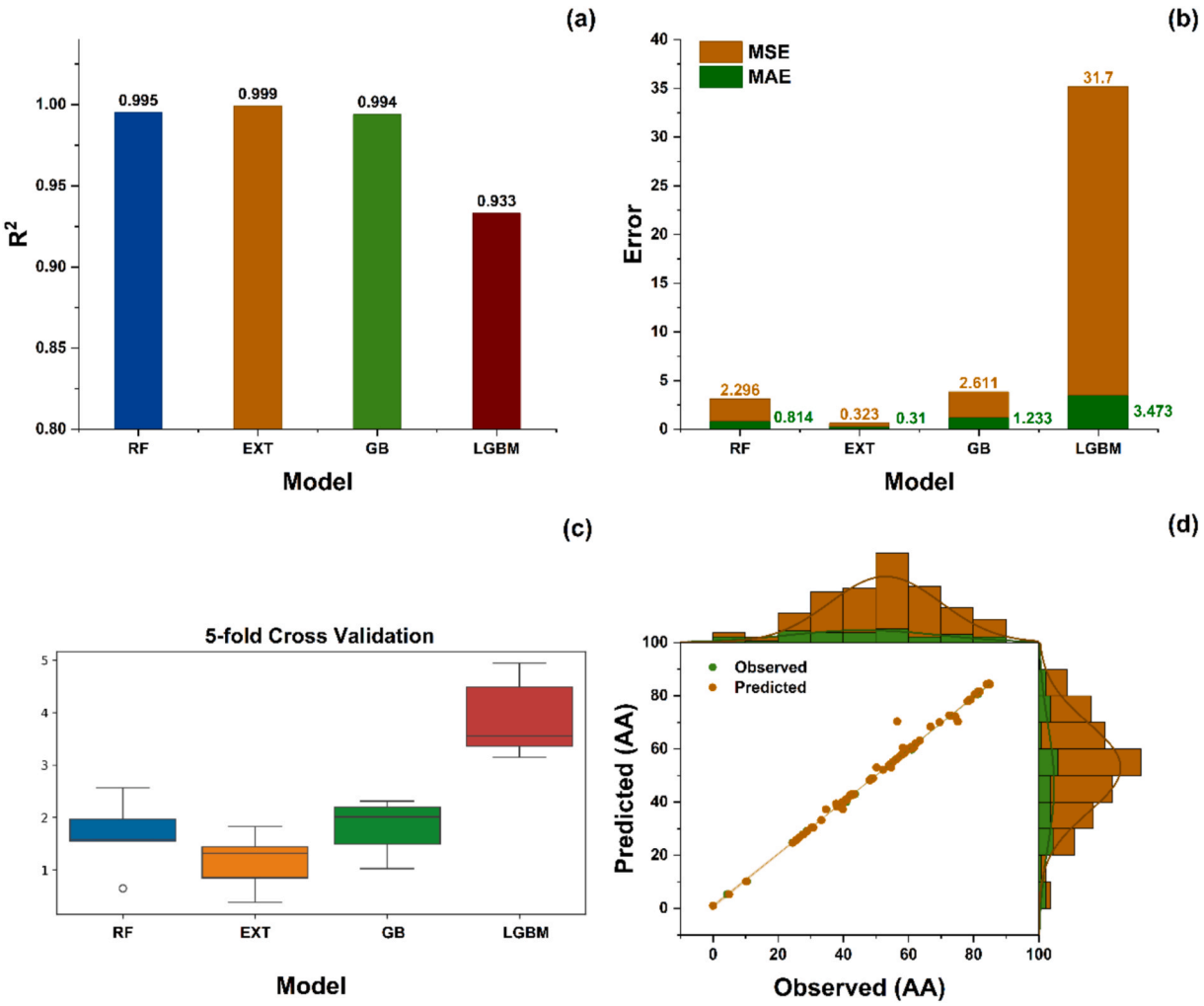



**Fig. 9.** Machine learning results. (a) $R^2$ score per model. (b) MAE and MSE for each model, with EXT as the best model. (c) RMSE averages for five-fold cross-validation. (d) Error plot for training and test datasets for the EXT model.

generated using the optimized EXT model (Fig. 9d). Finally, the importance of the attributes was evaluated using SHAP. SHAP values offer a more comprehensive method of feature selection by explaining the output of machine learning models based on cooperative game theory. SHAP values explain the contribution of each feature to individual predictions. SHAP analysis is usually used later in the process, when a trained model is available, and it is necessary to understand the impact of each feature on the model's predictions. The results of this analysis are shown in **Fig. S5**, where the variables are displayed on the y-axis and their importance is shown on the x-axis, as indicated by an increase in the MSE. According to the Shapley Indices, the most relevant feature is the ABTS method, followed by Concentration, and the $H_2O_2$ and DPPH methods. Other important variables are CS, ZP MLa, La, Mce, BG, and AWLa. Fig. 10 shows a violin plot of the SHAP Values, which visualizes the distribution and significance of feature contributions to the model's predictions. The width of each violin represents the density of SHAP values for a given feature, showing where the values are concentrated. Wider sections indicate where more data points have similar SHAP values, while narrower sections indicate less similar SHAP values. The position along the X-axis (or Y-axis, depending on orientation) shows the SHAP values, with positive values indicating an increase in the model's prediction and negative values indicating a decrease. By examining these distributions, it is possible to identify the features with the most substantial and consistent impact on the model's output.

### *3.3.4. Physical interpretation of ML results*

Considering the SHAP values presented in Fig. 10, the biological performance of the La-doped $CeO_2$ nanoparticles is on the result of several physical and chemical attributes that can be correlated with distinct antioxidant phenomena by means of the performed experiment, tested concentration, and availability of functional groups. The rationale behind these statements is addressed below.

a) Method employed and the importance of concentration for assessing antioxidant activity: A significant feature in the final model is the ABTS assay, followed by the $H_2O_2$ and DPPH techniques. This can be explained considering the wide reactivity of the ABTS reagent, which enhances its interaction with hydrophilic and lipophilic substances; this feature is also important when considering the hydrophilic nature of the synthesized nanoparticles due to the availability of carbonyl and hydroxyl functional groups and as $O^{2-}$ ions. As noted in Fig. 10, the $H_2O_2$ assay is another highly relevant main feature. This is because $H_2O_2$, which is a non-radical, is a reactive species that can be inhibited through a broad range of antioxidant mechanisms; this may make $CeO_2$ nanoparticles more effective due to their redox cycling capacity and mimetic intrinsic nature.

b) Influence of concentration: As shown in Fig. 10, concentration is another important feature because the biological performance of nanoparticles is based on dose–response relationships, where their antioxidant capacity increases with concentration up to a certain point, resulting in faster reactions and effective inhibition of free radicals.

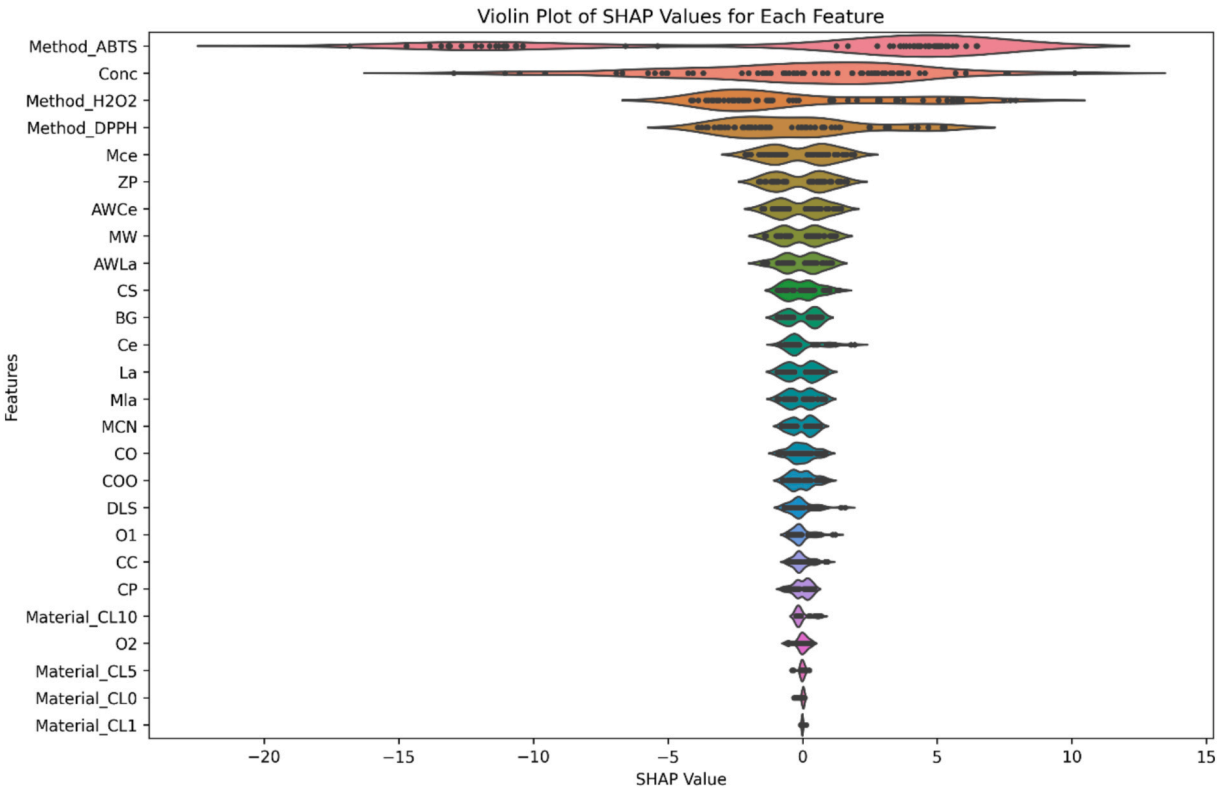



**Fig. 10.** Importance of each feature in the final model. The ABTS is the most important, followed by Concentration and Methods.

c) Availability of oxygen-based functional groups onto the surface of the La-doped $CeO_2$ nanoparticles: Recent scientific evidence suggests that the antioxidant activity of inorganic nanomaterials is due to their ability to transfer electrons to neutralize the formation of free radicals, such as DPPH and ABTS [55]. Similar findings have been reported for ceria-based nanomaterials, with a focus on their redox cycling capacity. However, researchers rarely consider that the scavenging activity of nanomaterials could be related to the existence of functional groups on their surfaces. Here, the developed nanomaterials exerted scavenging activity against DPPH, ABTS, and $H_2O_2$ in a dose-dependent manner, where the recorded activity was associated with the presence of carboxyl groups on the C0, C1, C5, and C10 surfaces. From a mechanistic point of view, carboxyl groups can neutralize the formation of free radicals through hydrogen atom donation and electron withdrawal. In the former mechanism, the donation of hydrogen atoms results in the formation of fewer reactive species. In contrast, in the latter, the negative charge of the carboxyl groups is stabilized through resonance, enhancing the reactivity of the donated hydrogen atom and its capability to neutralize the generation of free radicals. Unlike the electron-withdrawing capacity of other antioxidant functional groups, that of carboxyl groups can be modulated by exogenous or endogenous conditions that can enhance or inhibit their ability to counteract oxidative stress processes in biological systems [56]. This capacity of carboxyl groups should be considered, given that the antioxidant capacity of substances can depend on the polarity of solvents, where functional groups or engineered ligands can neutralize free radical formation through the formation of intramolecular hydrogen bonds [57]. In this regard, the carboxyl groups on the surfaces of C0, C1, C5, and C10 may have neutralized the generation of free radicals through intramolecular hydrogen bonding with the hydrazyl (–N–N–), nitro ($–NO_2$), and phenyl (aromatic rings) groups in DPPH. Interactions between these carboxyl groups and the sulfonic acid ($–SO_3H$) and aromatic groups in ABTS could have had a similar effect. The retrieved data were challenging to compare due to the limited evidence indicating the influence of functional groups or ligands on the antioxidant performance of ceria nanomaterials. Notably, only one study states that surface-engineered $CeO_{2-x}$ nanoparticles with *o*-quinone moieties exhibit high antioxidant capacity in biological environments containing DPPH and ROS [58]. Additionally, the zeta potential is a key factor in the stability of colloidal solutions, preventing agglomeration and thus providing more active sites for antioxidant reactions.

d) Oxygen vacancies. Consistent with Fig. 10 and according to Fig. 5a, the $CeO_2$ matrix also contains $O^{2-}$ ions and OH– groups, which can provide active oxygen vacancies. These vacancies are responsible for the redox cycling capacity of $CeO_2$, enabling it to convert potentially harmful free radical species into less reactive substances, thereby providing protection against oxidative damage in biological systems. A mechanistic approach is challenging to discuss since there are few studies on this subject. However, it can be hypothesized that, in the case of $O^{2-}$ ions, two unpaired electrons pairing in the same orbital can convert it into a highly reactive species capable of donating electrons and neutralizing the formation of DPPH, ABTS, and $H_2O_2$ radicals. Mechanistic insights into the activity of OH– groups in neutralizing the above-mentioned free radicals can be explained by the scavenging activity mechanisms of carboxyl groups, where electronegativity and solvation effects also influence performance.

## 4. Conclusions

This study successfully demonstrated the sonochemical synthesis of La-doped $CeO_2$ nanoparticles with varying lanthanum contents (0, 1, 5, and 10 at.%). Comprehensive characterization using XRD, DLS, TEM, XPS, and UV–vis spectroscopy confirmed the incorporation of La into the $CeO_2$ lattice, inducing changes in crystallinity, lattice parameters, and surface properties. The nanoparticles exhibited stable suspensions with controlled size distributions and low cytotoxicity across multiple cell lines (HepG-2, 3 T3-L1, Caco-2, and U87). Antioxidant assays (DPPH, ABTS, and $H_2O_2$) revealed enhanced free radical scavenging

capabilities, particularly in samples with 5 and 10 at.% La, showing superior antioxidant activity compared to undoped CeO2. This work's novelty lies in integrating advanced machine learning (ML) models—including Random Forest, Extremely Randomized Trees, Gradient Boosting, and LightGBM—to predict antioxidant activity based on physicochemical nanoparticle features. The ML models identified critical factors influencing antioxidant performance, such as the antioxidant assay method, nanoparticle concentration, zeta potential, surface carboxyl groups, and La content. Notably, the Extremely Randomized Trees model, after hyperparameter optimization, achieved outstanding predictive accuracy ($R^2$ close to 0.999), demonstrating the robust potential of ML in elucidating structure–activity relationships in nanomaterials. This combined experimental and data-driven approach provides mechanistic insights into how La doping modulates $CeO_2$ nanoparticle antioxidant behavior and highlights the significant promise of ML for the rational design and optimization of nanomaterials for biomedical applications. Future research will investigate the influence of dual-frequency, exposure duration, and various solvents on nanoparticle morphology and their biological effects.

## CRediT authorship contribution statement

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

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